



QUANTUM VIRTUAL INTERNSHIP

Retail Analytics

Customer & Chip Purchase Behaviour Analysis

246,740

Transactions

72,637

Customers

20

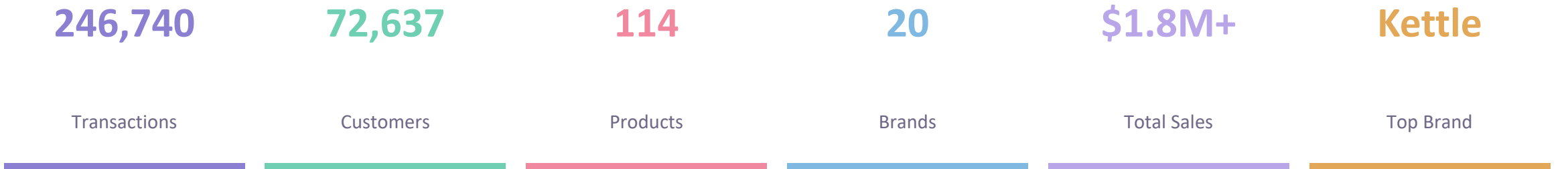
Brands

By Ishika Verma

THE NUMBERS AT A GLANCE

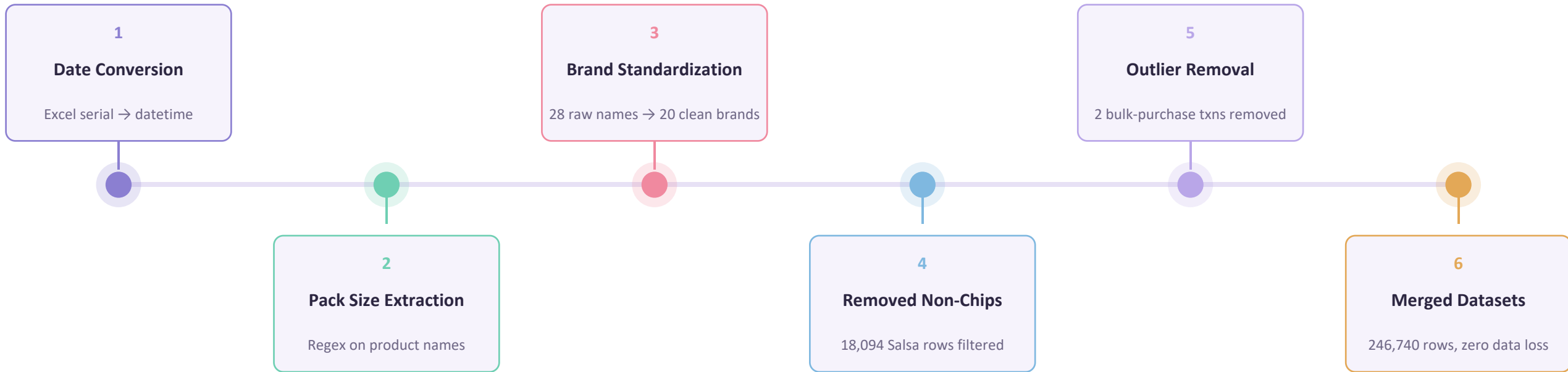
Project Overview

Objective: Analyze supermarket transaction and customer purchase behaviour data to identify valuable customer segments, high-performing brands, preferred pack sizes, and actionable business insights.



This project simulates a real Quantum retail analytics engagement — cleaning raw transaction data, engineering features, and translating patterns into commercial recommendations for a supermarket chain's chip category.

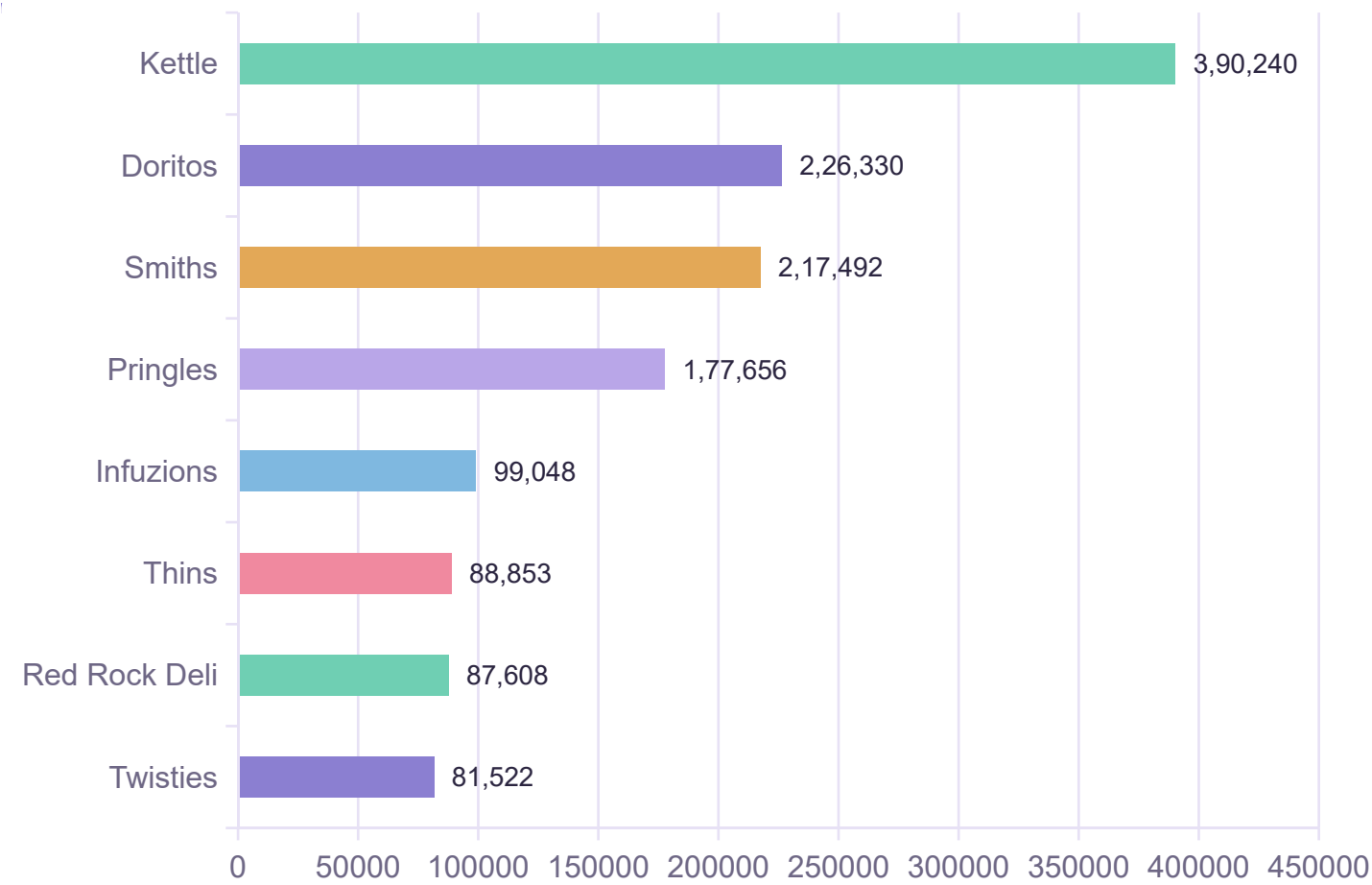
Data Cleaning & Feature Engineering



Six-step pipeline transformed 264,836 raw records into a clean, analysis-ready dataset of 246,740 transactions across 72,637 customers.

WHICH BRANDS DRIVE THE MOST REVENUE

Total Sales by Brand



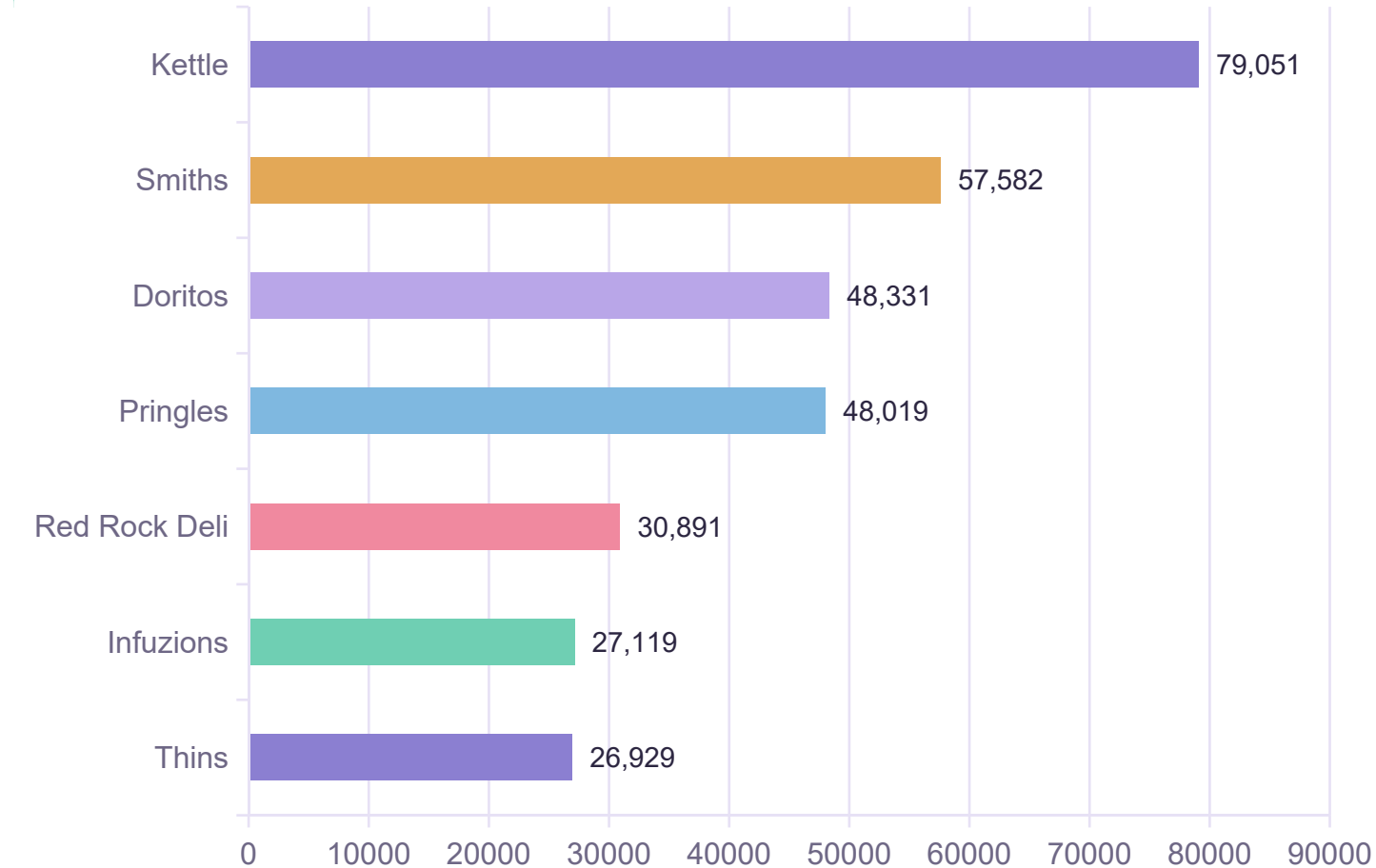
\$390,240

Kettle — Total Sales

Kettle leads by a wide margin — nearly 73% more than Doritos, the second-highest brand.

Kettle, Doritos, and Smiths together account for over 55% of total chip category revenue, making them the clear priority for shelf space and promotions.

Total Quantity Sold by Brand



79,051

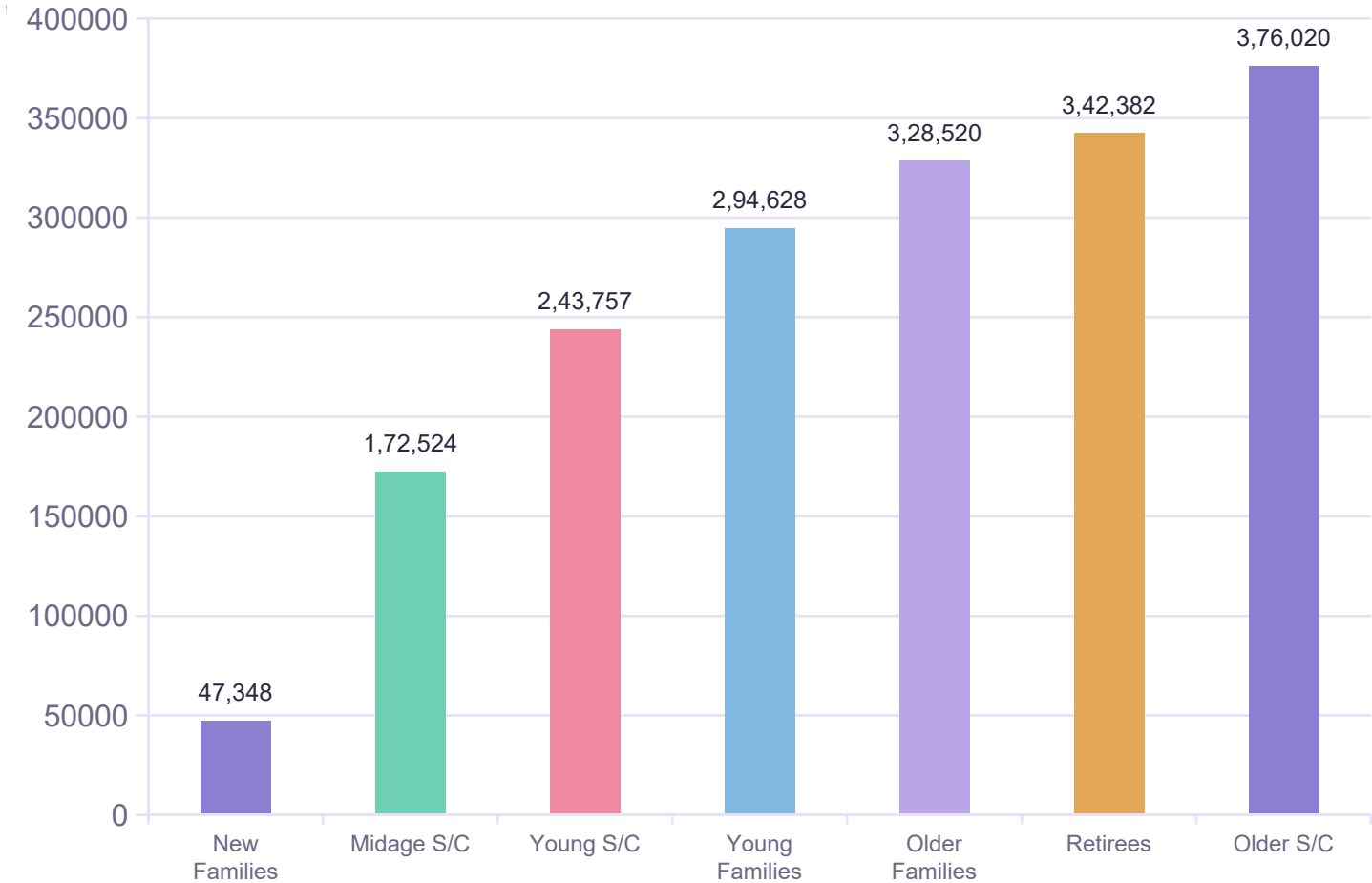
Kettle — Units Sold

Kettle sold the highest volume of any brand, confirming its lead is driven by genuine demand, not just higher pricing.

The sales-quantity pattern closely tracks total sales ranking, so stock planning should follow the same brand priority order.

WHICH CUSTOMER SEGMENTS SPEND THE MOST

Total Sales by Customer Lifestage



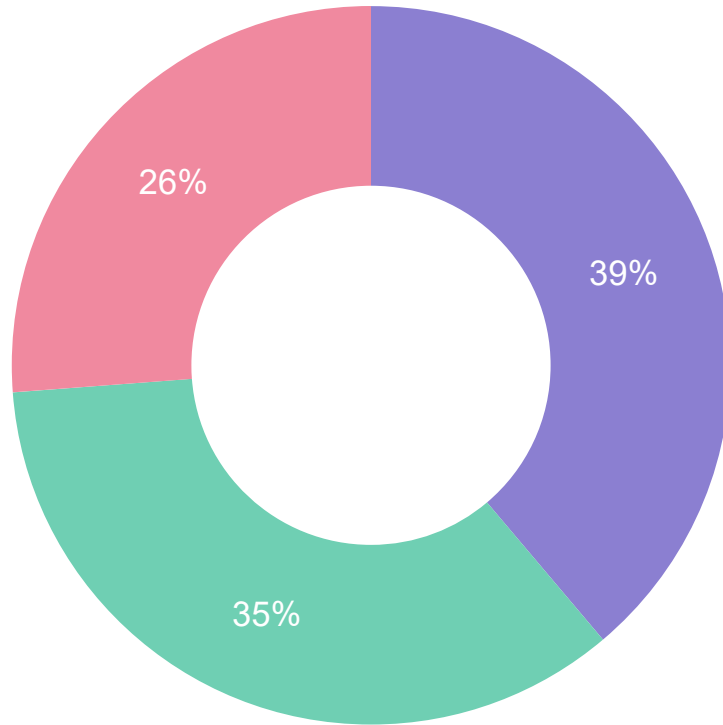
\$376,020

Older Singles/Couples

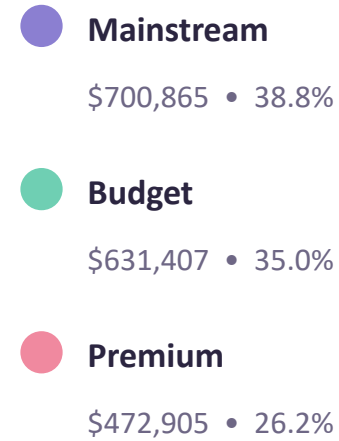
This segment generated the highest revenue of any lifestage, followed closely by Retirees and Older Families.

New Families contribute the least (\$47,348), suggesting an opportunity to build engagement with this segment through targeted offers.

Sales by Premium Customer Type



■ Mainstream ■ Budget ■ Premium



Mainstream customers lead revenue, narrowly ahead of Budget. Premium customers contribute the least (26.2%) despite typically higher spending power — an opportunity for premium-targeted bundles or promotions.

WHERE REVENUE REALLY COMES FROM

Sales by Lifestage & Premium Segment

	Budget	Mainstream	Premium
Midage Singles/Couples	\$33,346	\$84,734	\$54,444
New Families	\$20,607	\$15,980	\$10,761
Older Families	\$156,864	\$96,414	\$75,243
Older Singles/Couples	\$127,834	\$124,649	\$123,538
Retirees	\$105,916	\$145,169	\$91,297
Young Families	\$129,718	\$86,338	\$78,572
Young Singles/Couples	\$57,122	\$147,582	\$39,052

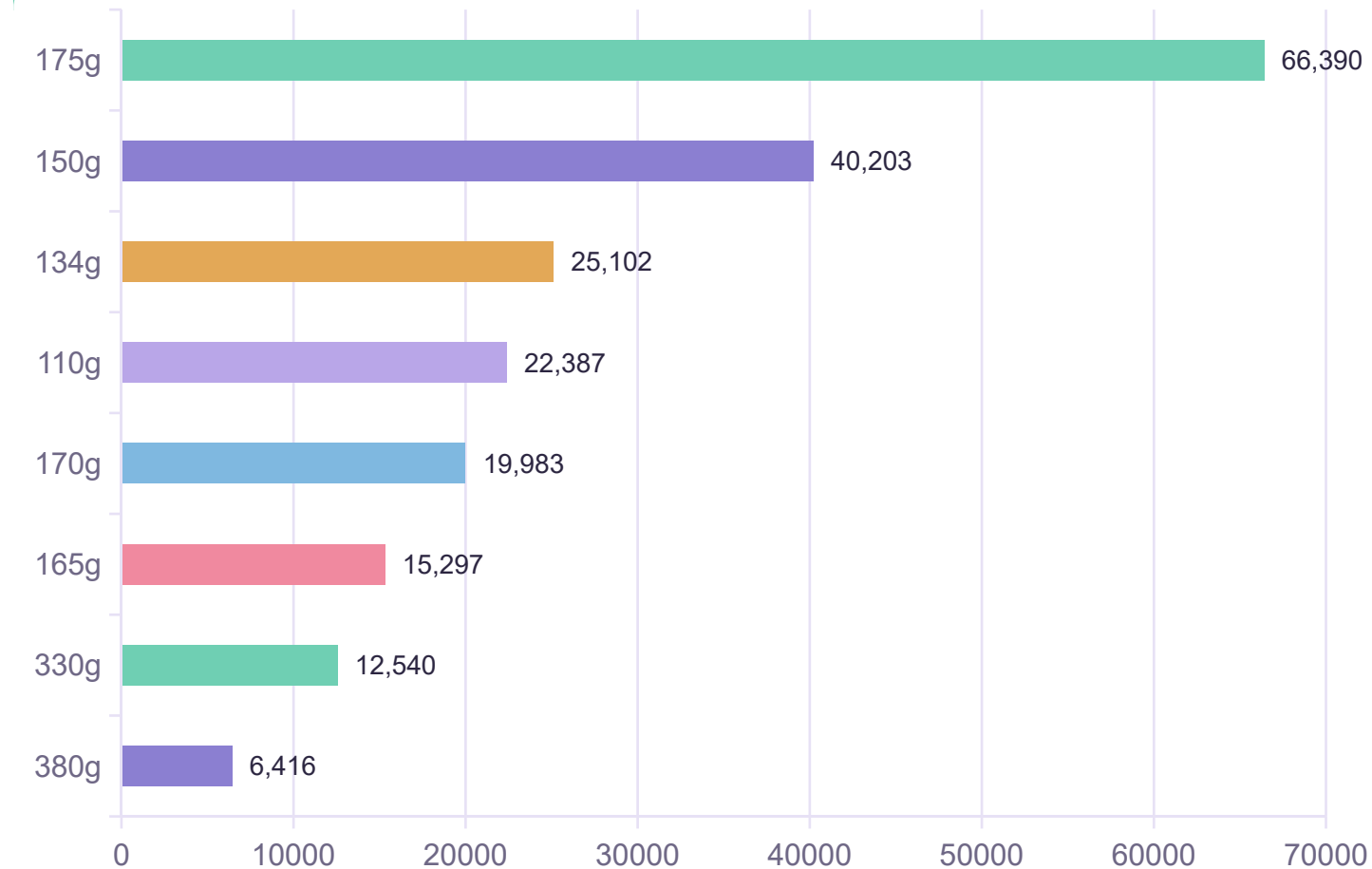
Three Standout Segments

Older Families (Budget): \$156,864
 Young Singles/Couples (Mainstream): \$147,582
 Older Singles/Couples (Premium): \$123,538

Each is the top performer within their own premium tier — spending depends on lifestage AND premium tier together.

WHICH SIZES CUSTOMERS ACTUALLY BUY

Pack Size Distribution



66,390

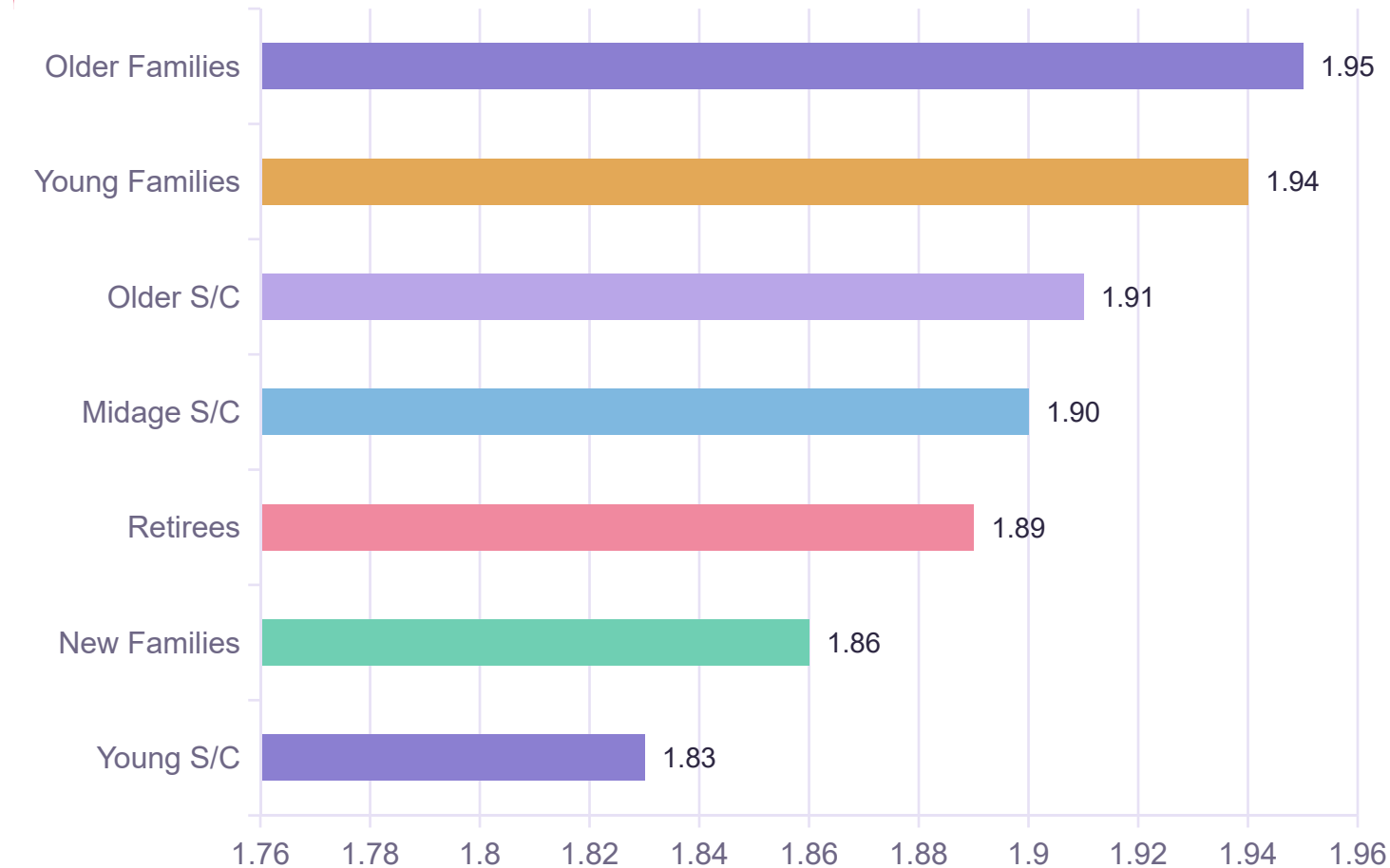
175g — Transactions

The 175g pack is the clear favorite — more than the next two sizes (150g, 134g) combined.

These top 3 sizes should be prioritized for shelf space and inventory, while very small or oversized packs may work better as promotional items.

PURCHASE BEHAVIOUR BY LIFESTAGE

Average Packs Purchased per Transaction



1.95

Older Families — Avg Packs

Older Families purchase the most packs per transaction on average, closely followed by Young Families at 1.94.

The spread is small, but family-oriented groups consistently buy marginally more, supporting multi-pack and family-sized promotions.

Key Business Insights

1

Kettle, Doritos, Smiths & Pringles are the top-performing brands by both sales and volume.

2

Older Singles/Couples, Retirees & Older Families generate the highest overall revenue.

3

Mainstream customers contribute the largest share of total sales (38.8%).

4

175g is the most popular pack size, followed by 150g and 134g.

5

Family-oriented segments purchase slightly larger quantities per transaction.

6

Spending patterns depend on both lifestage AND premium tier together.

CONCLUSION

Recommendations & Tools Used

Business Recommendations

- Prioritize inventory & shelf space for Kettle, Doritos, Smiths & Pringles
- Stock more of the 175g, 150g & 134g pack sizes
- Target Older Singles/Couples, Retirees & Older Families with loyalty campaigns
- Design premium-tier bundles to grow the underperforming Premium segment
- Use multi-pack promotions for family-oriented customer segments

Tools & Techniques Used

- Python
- Pandas & NumPy
- Regex (Feature Extraction)
- Seaborn & Matplotlib
- Jupyter Notebook
- Data Cleaning & EDA

Completed as part of the Quantum Virtual Internship program (Retail Analytics track).